DATASET INTELLIGENCE & OPERATIONAL ANALYSIS

Domain: Universal Dataset Intelligence & Distribution Analysis (0.0%) | Profiling Conf: 94.0% | Records: 450 | Hash: c6d4c2d5

Filter: ALL_RECORDS(UNFILTERED) | Audit: VERIFIED_DETERMINISTIC_PARITY | Engine: v8-javascript

KEY PERFORMANCE INDICATORS (DETERMINISTIC METRIC AUDIT)

TOTAL RECORDS VOLUME

450

`COUNTROWS`

Omitted: 0 rows

UNIQUE ENTITIES (PATIENT_ID)

450

`COUNTDISTINCT(Patient_ID)`

Omitted: 0 rows

TOTAL TOTAL_BILLED_USD

8,574,775

`SUM(Total_Billed_USD)`

Omitted: 0 rows

MEAN TOTAL_BILLED_USD

19,055.06

`AVERAGE(Total_Billed_USD)`

Omitted: 0 rows

MEDIAN TOTAL_BILLED_USD

18,229.50

`MEDIAN(Total_Billed_USD)`

Omitted: 0 rows

MEAN READMISSION_RISK_SCORE

5.50

`AVERAGE(Readmission_Risk_Score)`

Omitted: 0 rows

MEDIAN READMISSION_RISK_SCORE

5.50

`MEDIAN(Readmission_Risk_Score)`

Omitted: 0 rows

EXECUTIVE MANAGEMENT DECISION BRIEF

- **Ingestion & Data Quality Profile:** Audited 450 source rows across 7 columns with 100.0% full record completeness. Detected 0 duplicate records in source data.
- **Total Records Volume Benchmark:** Total Records Volume stands at 450 calculated via COUNTROWS. Supporting benchmark Mean Total_Billed_USD stands at 19,055.06.
- **Cohort Concentration (Department, Ward):** Largest cohort resides in 'Orthopedics' with 14,48,050 (16.9% of cohort total). Lead cohort exceeds 'Cardiology' by 1.0x.
- **Temporal Trajectory (Month):** Measured from 2024-01-01 (7,26,664) to 2024-12-01 (7,13,751) with -1.8% net trajectory. Peak volume recorded in 2024-05-01 (7,47,212).

DATASET INTEGRITY & STATISTICAL COVERAGE

Audited Records: 450 | Fully Populated: 450 (100.0%) | Duplicates: 0 | Missingness Exclusions Documented

VISUAL COHORT BREAKDOWNS & OPERATIONAL ACTION QUEUE

DISTRIBUTION BY DEPARTMENT_WARD (Department_Ward vs Total Total_Billed_USD)

ACTIVITY TREND OVER ADMISSION_DATE (Month vs Total Total_Billed_USD)

METHODOLOGY, AUDIT TRAIL & METRIC DICTIONARY

METHODOLOGICAL POLICIES & AGGREGATION AUDIT TRAIL

Policy #1: All analytical aggregations are computed locally in-browser using full-precision raw IEEE 754 float values.

Policy #2: No intermediate rounded values are ever re-aggregated across groups, metrics, or export layers.

Policy #3: Non-additive metrics (ratings, GPA, percentages, unit prices) are aggregated via weighted or arithmetic mean/median only; summation is strictly forbidden.

Policy #4: Semi-additive metrics (stock on hand, account balances) are point-in-time state indicators and must not be aggregated over time dimensions.

Policy #5: Missing, null, or empty string values are excluded on a per-metric basis without corrupting adjacent columns.

Policy #6: Zero AI hallucination guarantee: 100% deterministic code compilation with verified three-way engine parity.

CANONICAL METRIC DICTIONARY & ADDITIVITY TIERS

METRIC LABEL	FORMULA SPECIFICATION	ADDITIVITY	EXCLUSIONS	CANONICAL VALUE
Total Records Volume	`COUNTROWS`	**ADDITIVE**	0 omitted	**450**
Unique Entities (Patient_ID)	`COUNTDISTINCT(Patient_ID)`	**ADDITIVE**	0 omitted	**450**
Total Total_Billed_USD	`SUM(Total_Billed_USD)`	**ADDITIVE**	0 omitted	**8,574,775**
Mean Total_Billed_USD	`AVERAGE(Total_Billed_USD)`	**NON_ADDITIVE**	0 omitted	**19,055.06**
Median Total_Billed_USD	`MEDIAN(Total_Billed_USD)`	**ADDITIVE**	0 omitted	**18,229.50**
Mean Readmission_Risk_Score	`AVERAGE(Readmission_Risk_Score)`	**NON_ADDITIVE**	0 omitted	**5.50**
Median Readmission_Risk_Score	`MEDIAN(Readmission_Risk_Score)`	**ADDITIVE**	0 omitted	**5.50**

INDEPENDENT MATHEMATICAL RECONCILIATION & REPRODUCIBILITY GUARANTEE

Source Hash: c6d4c2d5 | Report Hash: 43ffe56b | Audit Timestamp: 2026-10-23T16:31:01.321Z

All metrics verified against independent Python float64 recalculation. Full raw precision preserved across all export targets.